

Ge158 Lecture 4: EM Wave Transmission and Reflection at Planar Interfaces, Part 1: Normal Incidence

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Topics covered

- Transverse waves on strings as an analog for EM waves
- Boundary conditions for transverse waves
- EM waves with normal incidence to a plane
- EM boundary conditions
- Fresnel equations for normal incidence

1 Transverse waves on a string

Let's start with some intuition building by considering the canonical example of waves on strings.

1.1 Longitudinal and transverse waves

There are two types of waves:

- *longitudinal waves* (e.g., sound waves, stretching of a Slinky, etc.);
- *transverse waves* (e.g., waves generated by shaking or plucking strings).

↪ EM (plane) waves are *always* transverse.

Recall the wave equation for a wave traveling in the z direction,

$$\frac{\partial^2 \xi}{\partial z^2} = \frac{1}{u_p^2} \frac{\partial^2 \xi}{\partial t^2}, \quad (1)$$

where $\xi(z, t)$ is the (transverse) displacement of the string. For a string, you may recall from physics classes that

$$u_p = \sqrt{\frac{F_{\text{tension}}}{m_\ell}}, \quad (2)$$

where F_{tension} is the tension of the string and m_ℓ is its mass per unit length. (For EM waves, we showed before that $u_p = c/n'$, where $n' = \text{Re}(\sqrt{\epsilon})$; Lecture 2.)

1.2 Two strings joined at $z = 0$

Let us consider two strings attached at $z = 0$: string 1 extends to $z \rightarrow -\infty$ and string 2 to $z \rightarrow +\infty$.

- $F_{\text{tension},1} = F_{\text{tension},2}$;
- $m_{\ell 1} \neq m_{\ell 2} \Rightarrow u_{p1} \neq u_{p2}$ (just as we'd expect for EM waves when $\varepsilon_1 \neq \varepsilon_2$).

An incident wave traveling in the $+z$ direction in string 1,

$$\xi^i = \xi_0^i e^{j(\omega t - k_1 z)} \quad (\text{string 1, } z < 0), \quad (3)$$

gives a reflected wave (traveling in the $-z$ direction in string 1)

$$\xi^r = \xi_0^r e^{j(\omega t + k_1 z)} \quad (\text{string 1, } z < 0) \quad (4)$$

and a transmitted wave (traveling in the $+z$ direction in string 2)

$$\xi^t = \xi_0^t e^{j(\omega t - k_2 z)} \quad (\text{string 2, } z > 0). \quad (5)$$

- The frequency ω must be the same everywhere and is set by the source (e.g., the person shaking the string, the RS instrument, etc.).
- Recalling the dispersion relation, $\omega = k u_p$: if $m_{\ell 1} \neq m_{\ell 2}$, then $u_{p1} \neq u_{p2}$ and $k_1 \neq k_2$.
 $\Rightarrow$ the wavelength must differ if $m_{\ell 1} \neq m_{\ell 2}$.

Now let us define the net displacement in each string:

$$\xi(z, t) = \begin{cases} \xi_0^i e^{j(\omega t - k_1 z)} + \xi_0^r e^{j(\omega t + k_1 z)} & \text{for } z < 0, \\ \xi_0^t e^{j(\omega t - k_2 z)} & \text{for } z > 0. \end{cases} \quad (6)$$

1.3 Boundary conditions and reflection and transmission

Define the boundary conditions (BCs) at $z = 0$:

1. $\xi(z = 0^-, t) = \xi(z = 0^+, t)$; otherwise, the strings would be disconnected.
2. $\left. \frac{\partial \xi}{\partial z} \right|_{0^-} = \left. \frac{\partial \xi}{\partial z} \right|_{0^+}$; otherwise, there would need to be a net force acting at $z = 0$.

Applying the BCs to Eq. (6), we have

$$\xi_0^i + \xi_0^r = \xi_0^t \quad \text{and} \quad k_1(\xi_0^i - \xi_0^r) = k_2 \xi_0^t. \quad (7)$$

In RS, we often know ξ_0^i , so we rearrange to get

$$\xi_0^r = \left(\frac{k_1 - k_2}{k_1 + k_2} \right) \xi_0^i = \left(\frac{u_{p2} - u_{p1}}{u_{p2} + u_{p1}} \right) \xi_0^i, \quad (8)$$

$$\xi_0^t = \left(\frac{2k_1}{k_1 + k_2} \right) \xi_0^i = \left(\frac{2u_{p2}}{u_{p2} + u_{p1}} \right) \xi_0^i. \quad (9)$$

As we went over with EM waves, ξ_0^i , ξ_0^r , and ξ_0^t can be complex-valued, and we can write them in terms of magnitude and phase (ϕ^i , ϕ^r , ϕ^t) as

$$|\xi_0^r| e^{j\phi^r} = \left(\frac{u_{p2} - u_{p1}}{u_{p2} + u_{p1}} \right) |\xi_0^i| e^{j\phi^i}, \quad |\xi_0^t| e^{j\phi^t} = \left(\frac{2u_{p2}}{u_{p2} + u_{p1}} \right) |\xi_0^i| e^{j\phi^i}. \quad (10)$$

Important take-aways from Eqs. (10):

- Continuity requires $\phi^t = \phi^i$.
- But Eqs. (10) require

$$\phi^r = \begin{cases} \phi^i - \pi & \text{for } u_{p1} > u_{p2} \quad (\text{reflected wave upside down}), \\ \phi^i & \text{for } u_{p2} > u_{p1} \quad (\text{reflected wave right side up}). \end{cases} \quad (11)$$

- If $m_{\ell 2} \gg m_{\ell 1}$ (similarly, if string 2 is fixed in place or, in the EM case, $|\varepsilon_2| \gg |\varepsilon_1|$, e.g., $\sigma_2/(\omega\varepsilon_0) \gg \varepsilon'_1$), then

$$u_{p2} \ll u_{p1} \quad \Rightarrow \quad |\xi_0^r| \approx |\xi_0^i| \quad \text{and} \quad |\xi_0^t| \approx 0. \quad (12)$$

2 EM boundary conditions

Having developed our intuition using the string analogy, let's move to EM waves. First, let's derive the BCs.

2.1 Maxwell's equations in integral form

Gauss's laws are integrals over any closed surface $\partial\mathcal{V}$ (the boundary of a volume $\mathcal{V}$) with normal $\hat{\mathbf{a}}$ (positive pointing from medium 2 to medium 1), where the circle on the integral sign denotes a *closed* surface:

$$\oint_{\partial\mathcal{V}} \varepsilon' \varepsilon_0 \mathbf{E} \cdot d\mathbf{a} = 0 \quad (\text{Gauss's law}), \quad (13)$$

$$\oint_{\partial\mathcal{V}} \mu \mathbf{H} \cdot d\mathbf{a} = 0 \quad (\text{Gauss's law for magnetism}). \quad (14)$$

Faraday's and Ampère's laws hold for any open surface $\mathcal{A}$ bounded by a closed loop $\partial\mathcal{A}$:

$$\oint_{\partial\mathcal{A}} \mathbf{E} \cdot d\boldsymbol{\ell} = -\mu \int_{\mathcal{A}} \frac{\partial \mathbf{H}}{\partial t} \cdot d\mathbf{a} \quad (\text{Faraday's law}), \quad (15)$$

$$\oint_{\partial\mathcal{A}} \mathbf{H} \cdot d\boldsymbol{\ell} = I_f + \varepsilon' \varepsilon_0 \int_{\mathcal{A}} \frac{\partial \mathbf{E}}{\partial t} \cdot d\mathbf{a} \quad (\text{Ampère's law}), \quad (16)$$

where I_f is the free current passing through the loop, taken to be $I_f = 0$ hereafter.

2.2 Gauss's law

Let's consider a very thin pillbox centered on the interface between two media and extending slightly into each (Figure 1).

From Gauss's law, we have

$$\varepsilon'_1 \mathbf{E}_1 \cdot \hat{\mathbf{a}} = \varepsilon'_2 \mathbf{E}_2 \cdot \hat{\mathbf{a}} \quad (17)$$

(note that the edges of our pillbox contribute nothing as the thickness of the box $\rightarrow 0$). Since $\hat{\mathbf{a}}$ is the normal,

$$\boxed{\varepsilon'_1 E_1^\perp = \varepsilon'_2 E_2^\perp} \quad (18)$$

where $E_1^\perp$ and $E_2^\perp$ denote the components of the $\mathbf{E}_1$ and $\mathbf{E}_2$ fields perpendicular to the interface.

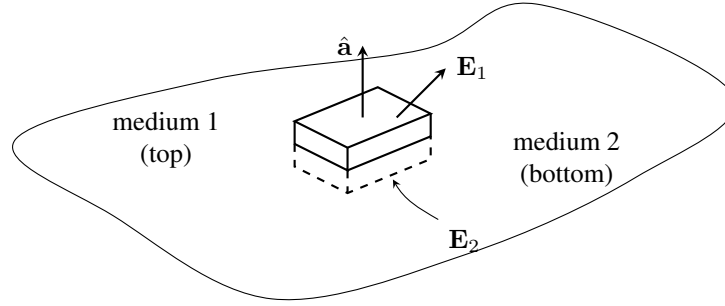


Figure 1: A thin pillbox straddling the interface between medium 1 (top) and medium 2 (bottom), with normal $\hat{\mathbf{a}}$. The part of the box in medium 2 is dashed.

2.3 Gauss's law for magnetism

Following similar logic as for Gauss's law for electricity, we have

$$\mu_1 H_1^\perp = \mu_2 H_2^\perp. \quad (19)$$

Given that in RS we nearly always have $\mu = \mu_0$, we can simplify to

$$\boxed{H_1^\perp = H_2^\perp} \quad (20)$$

$\Rightarrow$ the normal component of the magnetic field is conserved.

2.4 Faraday's law

Now imagine our pillbox is very narrow in one horizontal direction, and the long side has length ℓ , so that its cross-section is a thin rectangular loop straddling the interface. We have

$$(\mathbf{E}_1 - \mathbf{E}_2) \cdot \ell = -\frac{d}{dt} \int_{\mathcal{A}} \mu \mathbf{H} \cdot d\mathbf{a}. \quad (21)$$

As the height (thickness) of the loop $\rightarrow 0$, the area of $\mathcal{A}$ and hence the flux (right-hand side) go to 0:

$$\boxed{E_1^\parallel = E_2^\parallel} \quad (22)$$

$\Rightarrow$ the tangential component of the electric field is continuous across the interface.

★ *If you remember nothing else from this lecture, remember this BC, as it is crucial in RS applications and will appear over and over.*

2.5 Ampère's law

Following the same logic as for Faraday's law,

$$\boxed{H_1^\parallel = H_2^\parallel} \quad (\text{zero free current at the surface}) \quad (23)$$

$\Rightarrow$ the tangential component of the $\mathbf{H}$ field is conserved.

2.6 Summary

For RS applications (no free charge) and lossless materials, there are four BCs:

1. $\varepsilon'_1 E_1^\perp = \varepsilon'_2 E_2^\perp \rightarrow \varepsilon' \varepsilon_0 \mathbf{E}$ is the *electric displacement* $\Rightarrow$ the displacement normal to the interface is the same on both sides of the interface.
2. $H_1^\perp = H_2^\perp \rightarrow$ the normal component of the $\mathbf{H}$ field is the same on both sides of the interface (assuming $\mu_1 = \mu_2$, common in RS as $\mu = \mu_0$ in nearly all cases).
3. ★ $E_1^\parallel = E_2^\parallel \rightarrow$ the tangential component of the $\mathbf{E}$ field is conserved across the interface.
4. $H_1^\parallel = H_2^\parallel \rightarrow$ the tangential component of the $\mathbf{H}$ field is conserved across the interface.

3 EM wave reflection and transmission at normal incidence

3.1 Incident, reflected, and transmitted fields

Let's consider a simple and very useful example of normal incidence to a planar interface.

- We have a uniform plane wave initially propagating downward, $\hat{\mathbf{k}}_i = -\hat{\mathbf{z}}$.
- The interface between the two media is at $z = 0$ (Figure 2).
- The $\mathbf{E}$ field is polarized in the y -direction $\Rightarrow \mathbf{H}$ is in the x -direction.

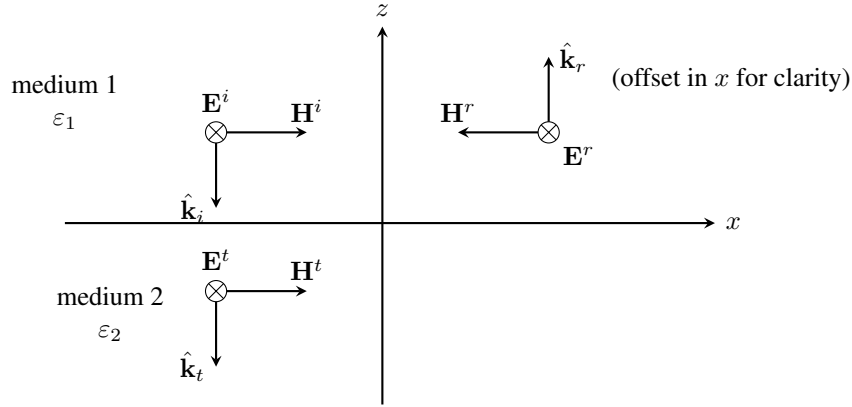


Figure 2: A uniform plane wave normally incident on the planar interface ($z = 0$) between medium 1 and medium 2. The $\mathbf{E}$ fields point along $+\hat{\mathbf{y}}$ (into the page). The incident, reflected, and transmitted waves are offset in x for clarity.

We define the fields (phasors) as

$$\text{medium 1, incident: } \tilde{\mathbf{E}}^i = \hat{\mathbf{y}} E^i e^{\gamma_1 z}, \quad \tilde{\mathbf{H}}^i = \hat{\mathbf{x}} \frac{E^i}{\eta_1} e^{\gamma_1 z}, \quad (24)$$

$$\text{medium 1, reflected: } \tilde{\mathbf{E}}^r = \hat{\mathbf{y}} E^r e^{-\gamma_1 z}, \quad \tilde{\mathbf{H}}^r = -\hat{\mathbf{x}} \frac{E^r}{\eta_1} e^{-\gamma_1 z}, \quad (25)$$

$$\text{medium 2, transmitted: } \tilde{\mathbf{E}}^t = \hat{\mathbf{y}} E^t e^{\gamma_2 z}, \quad \tilde{\mathbf{H}}^t = \hat{\mathbf{x}} \frac{E^t}{\eta_2} e^{\gamma_2 z}, \quad (26)$$

where (Lecture 2)

$$\gamma_1 = \alpha_1 + j\beta_1, \quad \eta_1 = \frac{\eta_0}{\sqrt{\epsilon_1}}, \quad \gamma_2 = \alpha_2 + j\beta_2, \quad \eta_2 = \frac{\eta_0}{\sqrt{\epsilon_2}}. \quad (27)$$

The total fields in each medium are

$$\tilde{\mathbf{E}}_1 = \tilde{\mathbf{E}}^i + \tilde{\mathbf{E}}^r, \quad \tilde{\mathbf{H}}_1 = \tilde{\mathbf{H}}^i + \tilde{\mathbf{H}}^r, \quad \tilde{\mathbf{E}}_2 = \tilde{\mathbf{E}}^t, \quad \tilde{\mathbf{H}}_2 = \tilde{\mathbf{H}}^t. \quad (28)$$

3.2 Boundary conditions and the Fresnel coefficients

- At $z = 0$, we must apply the BCs from Maxwell's equations.
- At normal incidence, the fields are entirely tangential to the interface (again, because EM plane waves are always transverse), so

$$E^i + E^r = E^t \quad \rightarrow \text{tangential component of the } \mathbf{E} \text{ field conserved}, \quad (29)$$

$$\frac{E^i}{\eta_1} - \frac{E^r}{\eta_1} = \frac{E^t}{\eta_2} \quad \rightarrow \text{tangential component of the } \mathbf{H} \text{ field conserved}. \quad (30)$$

- In general in RS, we know E^i , so we solve for the reflection and transmission coefficients

$$\rho = \frac{E^r}{E^i} \quad \text{and} \quad \tau = \frac{E^t}{E^i}. \quad (31)$$

- From above, we see

$$1 + \rho = \tau \quad \text{and} \quad 1 - \rho = \frac{\eta_1}{\eta_2} \tau. \quad (32)$$

- Solving gives the *Fresnel reflection* (ρ) and *transmission* (τ) coefficients for normal incidence:

$$\rho = \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{\sqrt{\epsilon_1} - \sqrt{\epsilon_2}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}}, \quad \tau = \frac{2\eta_2}{\eta_2 + \eta_1} = \frac{2\sqrt{\epsilon_1}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}} \quad (33)$$

3.3 Connecting to the string model

You may notice the similarity of forms between ρ and τ and reflection and transmission on strings. Connecting to the string model given at the beginning of the lecture, consider *lossless* media ($\epsilon_1 = \epsilon'_1$ and $\epsilon_2 = \epsilon'_2$) and recall that

$$u_{p1} = \frac{c}{\sqrt{\epsilon_1}} \quad \text{and} \quad u_{p2} = \frac{c}{\sqrt{\epsilon_2}}, \quad (34)$$

so

$$\rho = \frac{u_{p2} - u_{p1}}{u_{p2} + u_{p1}} = \frac{k_1 - k_2}{k_1 + k_2}, \quad \tau = \frac{2u_{p2}}{u_{p2} + u_{p1}} = \frac{2k_1}{k_1 + k_2}, \quad (35)$$

which are identical to the reflection and transmission coefficients on a string!

3.3.1 ★ Key take-away point

The reflected EM wave is

- in phase with the incident wave if

$$u_{p1} < u_{p2} \Rightarrow \operatorname{Re}(\sqrt{\varepsilon_1}) > \operatorname{Re}(\sqrt{\varepsilon_2}); \quad (36)$$

- out of phase (π offset) with the incident wave if

$$u_{p1} > u_{p2} \Rightarrow \operatorname{Re}(\sqrt{\varepsilon_1}) < \operatorname{Re}(\sqrt{\varepsilon_2}), \quad (37)$$

which is virtually always the case when medium 1 is the atmosphere.

- These conditions are exact for lossless media, where ρ is real and the phase offset is exactly 0 or π . For lossy media, ρ is complex and the offset lies in between. In general, $\operatorname{Re}(\rho) = (|\varepsilon_1| - |\varepsilon_2|) / |\sqrt{\varepsilon_1} + \sqrt{\varepsilon_2}|^2$, so the reflected wave is closer to in phase ($\operatorname{Re}(\rho) > 0$) when $|\varepsilon_1| > |\varepsilon_2|$ and closer to out of phase ($\operatorname{Re}(\rho) < 0$) when $|\varepsilon_1| < |\varepsilon_2|$. For air over water, ice, or soil, $|\varepsilon_2| \gg |\varepsilon_1|$, so the offset is near π .

3.4 Power: reflectivity and transmissivity

We may also want to think about this in terms of the partitioning of energy between the reflected and transmitted waves. Using the time-averaged Poynting vector (Lecture 2) and writing $\tilde{\mathbf{E}}^r = \hat{\mathbf{y}}\rho E^i e^{-\gamma_1 z}$, we have in medium 1

$$\begin{aligned} \mathcal{S}_{\text{av}1} &= \frac{1}{2} \operatorname{Re} [\tilde{\mathbf{E}}_1 \times \tilde{\mathbf{H}}_1^*] \\ &= \frac{1}{2} \operatorname{Re} \left[\hat{\mathbf{y}} E^i (e^{\gamma_1 z} + \rho e^{-\gamma_1 z}) \times \hat{\mathbf{x}} \frac{E^{i*}}{\eta_1} (e^{-\gamma_1 z} - \rho^* e^{\gamma_1 z}) \right], \end{aligned} \quad (38)$$

taking medium 1 to be lossless ($\gamma_1 = j\beta_1$, η_1 real; e.g., the atmosphere), while medium 2 may be lossy. At $z = 0$, this gives

$$\mathcal{S}_{\text{av}1} = -\hat{\mathbf{z}} \frac{|E^i|^2}{2|\eta_1|} \underbrace{(1 - |\rho|^2)}_{\text{note this term!}} \cos \phi_{\eta_1} = \mathcal{S}_{\text{av}}^i + \mathcal{S}_{\text{av}}^r, \quad (39)$$

the sum of the incident and reflected power densities. Similarly,

$$\mathcal{S}_{\text{av}}^t = \mathcal{S}_{\text{av}2} = \frac{1}{2} \operatorname{Re} [\tilde{\mathbf{E}}_2 \times \tilde{\mathbf{H}}_2^*] = -\hat{\mathbf{z}} |\tau|^2 \frac{|E^i|^2}{2|\eta_2|} \cos \phi_{\eta_2} \quad (40)$$

at $z = 0$, where ϕ_{η_1} and ϕ_{η_2} are the phase angles of η_1 and η_2 (Lecture 2). From these expressions, we can show

$$|\tau|^2 = \frac{|\eta_2| \cos \phi_{\eta_1}}{|\eta_1| \cos \phi_{\eta_2}} (1 - |\rho|^2) \quad (41)$$

and

$$\mathcal{S}_{\text{av}1} = \mathcal{S}_{\text{av}2} \quad (\text{sanity check: energy is conserved!}). \quad (42)$$

Finally, it is common to define the reflectivity Γ and transmissivity $\mathbb{T}$,

$$\Gamma = \frac{|\mathcal{S}_{\text{av}}^r|}{|\mathcal{S}_{\text{av}}^i|} \Big|_{z=0} = |\rho|^2 \quad \text{and} \quad \mathbb{T} = \frac{|\mathcal{S}_{\text{av}}^t|}{|\mathcal{S}_{\text{av}}^i|} \Big|_{z=0} = |\tau|^2 \frac{|\eta_1| \cos \phi_{\eta_2}}{|\eta_2| \cos \phi_{\eta_1}}, \quad (43)$$

where it can be shown that

$$\Gamma + \mathbb{T} = 1. \quad (44)$$

Variables

Units are SI. Values are given for universal constants; “typical” values are representative, not definitive.

Symbol	Description	Units	Value
<i>Universal constants</i>			
ε_0	permittivity of free space	F/m	8.854×10^{-12}
μ_0	permeability of free space	H/m	$4\pi \times 10^{-7} \approx 1.257 \times 10^{-6}$
c	speed of light in a vacuum, $1/\sqrt{\mu_0\varepsilon_0}$	m/s	2.998×10^8
η_0	intrinsic impedance of free space, $\sqrt{\mu_0/\varepsilon_0}$	Ω	$\approx 376.7 (\approx 120\pi)$
j	imaginary unit, $j^2 = -1$ (time dependence $e^{j\omega t}$)	–	–
<i>Constitutive parameters</i>			
ε	complex dielectric constant (relative), $\varepsilon' - j\varepsilon''$	dimensionless	–
ε'	relative permittivity (real part of ε)	dimensionless	≥ 1 (almost always)
$\varepsilon_1, \varepsilon_2$	dielectric constants of media 1 and 2	dimensionless	–
μ	magnetic permeability	H/m	$\approx \mu_0$ in RS
μ_1, μ_2	permeabilities of media 1 and 2	H/m	μ_0 in RS
σ_1, σ_2	conductivities of media 1 and 2	S/m	–
ω	angular frequency (set by the source)	rad/s	–
<i>Waves on strings</i>			
ξ	transverse displacement of the string	m	–
ξ^i, ξ^r, ξ^t	incident, reflected, and transmitted waves on the strings	m	–
$\xi_0^i, \xi_0^r, \xi_0^t$	complex amplitudes of ξ^i, ξ^r, ξ^t	m	–
ϕ^i, ϕ^r, ϕ^t	phases of $\xi_0^i, \xi_0^r, \xi_0^t$	rad	$\phi^t = \phi^i$
F_{tension}	string tension ($F_{\text{tension},1}, F_{\text{tension},2}$ for strings 1 and 2)	N	equal in both strings
$m_\ell, m_{\ell 1}, m_{\ell 2}$	mass per unit length of the string (of strings 1 and 2)	kg/m	–
u_p	phase velocity; on a string, $\sqrt{F_{\text{tension}}/m_\ell}$; EM waves, c/n'	m/s	–
u_{p1}, u_{p2}	phase velocities in strings (media) 1 and 2	m/s	lossless EM: $c/\sqrt{\varepsilon_{1,2}}$
k_1, k_2	angular wavenumbers in strings (media) 1 and 2, ω/u_p	rad/m	–
z	coordinate along the string; normal to the interface (EM)	m	interface at $z = 0$
t	time	s	–
n'	real part of the index of refraction, $\text{Re}(\sqrt{\varepsilon})$	dimensionless	–
<i>Boundary conditions</i>			
E, H	electric and magnetic field intensity vectors (instantaneous)	V/m; A/m	–
E₁, E₂	(total) electric field in media 1 and 2	V/m	–
H₁, H₂	(total) magnetic field in media 1 and 2	A/m	–

Symbol	Description	Units	Value
$E^\perp, H^\perp$	field components perpendicular (normal) to the interface	V/m; A/m	–
$E^\parallel, H^\parallel$	field components parallel (tangential) to the interface	V/m; A/m	–
$\mathcal{V}, \partial\mathcal{V}$	volume and its closed bounding surface (Gauss's laws)	$\text{m}^3; \text{m}^2$	–
$\mathcal{A}, \partial\mathcal{A}$	open surface of integration and the closed loop bounding it	$\text{m}^2; -$	–
$d\mathbf{a}$	vector area element	m^2	–
$\hat{\mathbf{a}}$	unit normal to the interface, from medium 2 to medium 1	dimensionless	–
$d\ell$	vector line element along $\partial\mathcal{A}$	m	–
$\ell, \boldsymbol{\ell}$	length (and vector) of the long side of the loop	m	–
I_f	free current through the loop	A	0 (assumed)
<i>Normal incidence</i>			
$\tilde{\mathbf{E}}^i, \tilde{\mathbf{E}}^r, \tilde{\mathbf{E}}^t$	phasors of the incident, reflected, and transmitted electric fields	V/m	–
$\tilde{\mathbf{H}}^i, \tilde{\mathbf{H}}^r, \tilde{\mathbf{H}}^t$	phasors of the incident, reflected, and transmitted magnetic fields	A/m	–
$\tilde{\mathbf{E}}_1, \tilde{\mathbf{H}}_1$	total field phasors in medium 1	V/m; A/m	–
$\tilde{\mathbf{E}}_2, \tilde{\mathbf{H}}_2$	total field phasors in medium 2	V/m; A/m	–
E^i, E^r, E^t	complex amplitudes of the incident, reflected, and transmitted $\mathbf{E}$ fields at $z = 0$	V/m	E^i known
$\hat{\mathbf{k}}_i, \hat{\mathbf{k}}_r, \hat{\mathbf{k}}_t$	propagation unit vectors of the incident, reflected, and transmitted waves	dimensionless	$-\hat{\mathbf{z}}, \hat{\mathbf{z}}, -\hat{\mathbf{z}}$
$\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}$	Cartesian unit vectors	dimensionless	–
γ_1, γ_2	propagation constants of media 1 and 2, $\alpha + j\beta$	m^{-1}	–
α_1, α_2	attenuation constants of media 1 and 2	Np/m	0 if lossless
β_1, β_2	phase constants of media 1 and 2	rad/m	$k_{1,2}$ if lossless
η_1, η_2	intrinsic impedances of media 1 and 2, $\eta_0/\sqrt{\epsilon_{1,2}}$	Ω	real if lossless
$\phi_{\eta_1}, \phi_{\eta_2}$	phase angles of η_1 and η_2	rad	0 if lossless
ρ	Fresnel reflection coefficient (normal incidence), E^r/E^i	dimensionless	$ \rho \leq 1$
τ	Fresnel transmission coefficient (normal incidence), E^t/E^i	dimensionless	$1 + \rho$
$\mathcal{S}_{\text{av}1}, \mathcal{S}_{\text{av}2}$	time-averaged power density (Poynting vector) in media 1 and 2	W/m^2	equal at $z = 0$
$\mathcal{S}_{\text{av}}^i, \mathcal{S}_{\text{av}}^r, \mathcal{S}_{\text{av}}^t$	time-averaged power densities of the incident, reflected, and transmitted waves	W/m^2	–
Γ	reflectivity, $ \rho ^2$	dimensionless	$0 \leq \Gamma \leq 1$
$\mathbb{T}$	transmissivity, $1 - \Gamma$ (medium 1 lossless)	dimensionless	$0 \leq \mathbb{T} \leq 1$